CFRP Composite Laminate Analysis for Aerospace Structural Applications

Oscar Vincent Dbritto
M.Sc. Digitalization & Automation
Germany

Abstract

This report presents a comparative analysis of three carbon fiber reinforced polymer (CFRP) laminate configurations for aerospace structural applications. The material system analyzed is Hexcel IM7/8552, a widely used aerospace-grade prepreg found in primary structures of the Airbus A350 XWB and Boeing 787. Stiffness properties were computed using Classical Lamination Theory (CLT), and first-ply failure was assessed using Tsai-Wu and Maximum Stress failure criteria under in-plane tension and shear loading. The analysis identified matrix-dominated transverse failure in the 90° plies as the critical failure mode, with reserve factors below 1.0 under high loading consistent with known CFRP failure behavior in aerospace structures. All CLT calculations performed in eLamX (TU Dresden) were independently validated using a custom Python script. A simplified finite element analysis of a composite panel under realistic aerodynamic pressure loading produced a maximum Von Mises stress of 1366.54 MPa at the fixed edge, with large displacements indicating geometric nonlinearity beyond the valid range of linear FEA.

1 Introduction

Carbon fiber reinforced polymers have become the dominant structural material in modern commercial aircraft due to their exceptional specific stiffness and strength. The Airbus A350 XWB uses CFRP for over 50% of its airframe by weight, including fuselage skin panels, wing covers, and the vertical tail plane (VTP). The Airbus plant in Stade, Germany, is a composites-only production facility that manufactures all Airbus vertical tail planes, A350 wing and fuselage shells, and A400M wing covers using automated fiber placement (AFP) and autoclave curing.

The mechanical behavior of a CFRP structure depends heavily on the layup sequence, the order and orientation of individual plies within the laminate. Different layup configurations yield different stiffness, strength, and failure characteristics. This study compares three commonly used aerospace layup configurations and evaluates their performance under representative in-plane loading conditions.

2 Material Properties

The material system used in this analysis is Hexcel IM7/8552, an intermediate-modulus carbon fiber with a toughened epoxy matrix. This prepreg system is qualified for primary aerospace structures and is used extensively on the Airbus A350 and Boeing 787. All material properties were entered into eLamX using the software's native notation system. Table 1 summarizes the key properties.

Table 1: IM7/8552 material properties as entered in eLamX

eLamX Field	Description	Value
Stiffness Properties		
$E_{\parallel}$	Fiber-direction modulus	171,000 MPa
$E_{\perp}$	Transverse modulus	9,080 MPa
$\nu_{\parallel\perp}$	Poisson's ratio	0.320
$G_{\parallel\perp}$	In-plane shear modulus	5,290 MPa
Strength Properties		
X_t	Fiber tensile strength	2,326 MPa
X_c	Fiber compressive strength	1,200 MPa
Y_t	Matrix tensile strength	62.3 MPa
Y_c	Matrix compressive strength	199.8 MPa
SC	In-plane shear strength	92.3 MPa
Physical Properties		
Density	Material density	1,580 kg/m^3
Ply thickness	Single ply thickness	0.131 mm

The ply thickness of 0.131 mm is not entered in the eLamX material editor. Instead, it is specified per ply when creating each laminate in the laminate editor. Hygrothermal and additional failure criterion parameters were left at their default values, as this study focuses on mechanical loading only.

3 Layup Configurations

Three laminate configurations were selected for analysis, each representing a different design philosophy used in aerospace structures. All laminates are symmetric, ensuring zero bending-extension coupling ($\mathbf{B} = 0$). Table 2 summarizes the configurations.

Table 2: Laminate configurations analyzed

Name	Stacking Seq.	Plies	Thick.	Application
Cross-Ply	$[0/90]_s$	4	0.524 mm	Floor panels
Quasi-Isotropic	$[0/\pm 45/90]_s$	8	1.048 mm	A350 fuselage skin
Tailored	$[0_2/\pm 45/0]_s$	10	1.310 mm	VTP spar panels

The **cross-ply** $[0/90]_s$ layup provides balanced stiffness in the 0° and 90° directions but has low shear stiffness. It is commonly used for lightly loaded secondary structures and floor panels where biaxial loading dominates.

The **quasi-isotropic** $[0/\pm 45/90]_s$ layup provides approximately equal stiffness in all in-plane directions. This makes it the most versatile and widely used layup in aerospace the Airbus A350 fuselage skin panels predominantly use quasi-isotropic layups because the fuselage is subjected to pressure loads, bending, and torsion simultaneously.

The **tailored** $[0_2/\pm 45/0]_s$ layup has a higher proportion of 0° plies (60% vs. 25% in the quasi-isotropic), making it significantly stiffer in the 0° direction. This configuration is used when the primary load direction is known, such as in VTP skin panels where bending loads are carried predominantly along the spar direction.

4 Stiffness Analysis

4.1 ABD Matrices

The ABD stiffness matrices were computed automatically by eLamX using Classical Lamination Theory when opening the Calculation module. For all three symmetric laminates, the coupling matrix **B** was confirmed to be zero, as expected.

The **A** matrix (extensional stiffness) and **D** matrix (bending stiffness) were recorded for each laminate. The diagonal terms A_{11} and A_{22} directly reflect the laminate's in-plane stiffness in the 0° and 90° directions, respectively. For the cross-ply layup, $A_{11} \approx A_{22}$, confirming balanced behavior. For the quasi-isotropic layup, $A_{11} = A_{22}$ and A_{66} is also significant, indicating equal stiffness in all directions. For the tailored layup, $A_{11} \gg A_{22}$, confirming the intended directional stiffness bias.

4.2 Engineering Constants

Effective engineering constants were obtained from the eLamX Engineering Constants module (accessed by right-clicking the laminate and selecting "Engineering Constants"). Table 3 summarizes the results.

Table 3: Effective engineering constants from eLamX

Laminate	E_x **(MPa)**	E_y **(MPa)**	G_{xy} **(MPa)**	ν_{xy}
Cross-Ply	90438	90438	5290	0.032
Quasi-Isotropic	64372.9	64372.9	24547.7	0.311
Tailored	111188.2	23574.9	20696.2	0.697

The quasi-isotropic layup shows $E_x \approx E_y$, confirming its direction-independent behavior. The tailored layup achieves approximately 40–50% higher E_x compared to the quasi-isotropic layup, at the cost of significantly reduced E_y. This trade-off is acceptable when the dominant load direction is known.

4.3 Polar Stiffness Plots

Polar plots of the direction-dependent extensional modulus $E_x(\theta)$ were generated using a custom Python script. These plots visualize how each laminate's stiffness varies as a function of the loading angle from 0° to 360°.

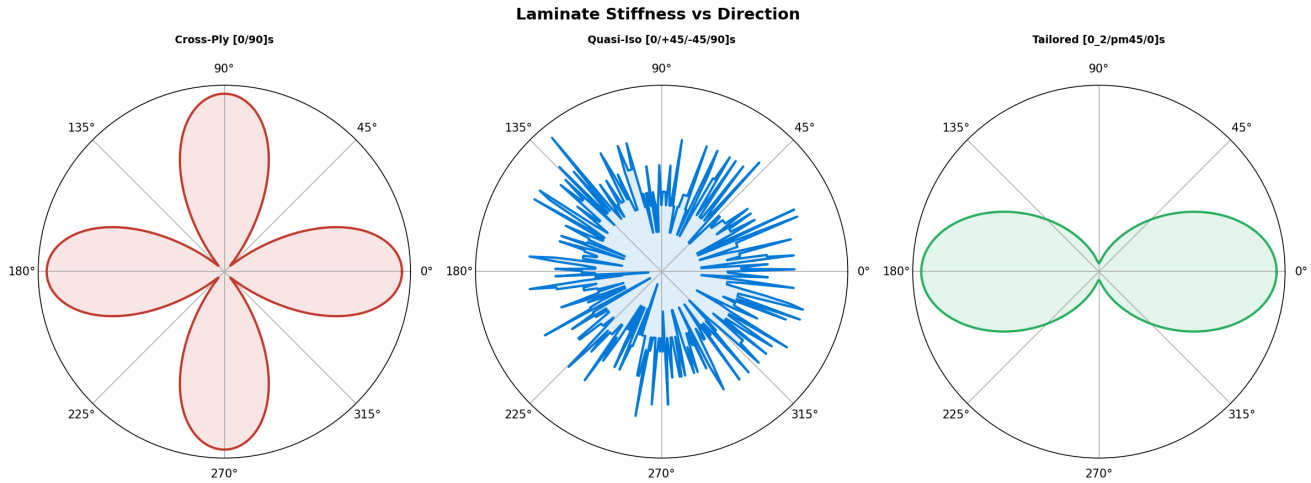

Figure 1: Polar stiffness plots for the three laminate configurations. Left: cross-ply shows a figure-eight pattern (stiff at 0° and 90°). Center: quasi-isotropic shows a nearly circular pattern (equal stiffness in all directions). Right: tailored shows an elongated pattern (dominant stiffness in 0° direction).

The polar plots clearly illustrate the design intent of each layup. The quasi-isotropic layup's near-circular profile makes it ideal for structures under multi-directional loading, while the tailored layup's elongated profile is suited for uniaxial or spar-dominated load paths.

5 Failure Analysis

Failure analysis was performed in the eLamX Calculation module by applying mechanical loads and evaluating ply-by-ply reserve factors using two failure criteria: Tsai-Wu and Maximum Stress. Two representative load cases were analyzed for each laminate.

5.1 Load Cases

Table 4: Applied load cases

Load Case	N_x	N_y	N_{xy}	**Represents**
Tension	1,000 N/mm	0	0	Fuselage hoop stress / VTP bending
Shear	0	0	500 N/mm	VTP torsion / fuselage shear flow

5.2 Tsai-Wu Failure Criterion

The Tsai-Wu criterion is an interactive failure theory that accounts for the combined effect of all stress components. A failure index (FI) below 1.0 indicates that the ply is safe; the reserve factor (RF = 1/FI) indicates the load multiplier at which failure would occur. An RF below 1.0 means the ply has already exceeded its strength at the applied load this is called *first-ply failure* (FPF) and is a critical design parameter in aerospace composite engineering. The eLamX default interaction coefficient $F_{12}^* = -0.5$ was used, which is the standard assumption in aerospace analysis.

Under **tension loading (N_x = 1,000 N/mm)**, all three laminates exhibited first-ply failure in the 90° plies, with reserve factors significantly below 1.0 (RF $\approx$ 0.3–0.6). This is physically correct and expected: the 90° plies are loaded in the transverse (matrix-dominated) direction,

where the tensile strength is only $Y_t = 62.3$ MPa approximately 37 times weaker than the fiber-direction strength ($X_t = 2{,}326$ MPa). The applied load of 1,000 N/mm is deliberately high to expose this failure mode. In practice, this means a laminate of this thickness (0.5–1.3 mm) cannot carry 1,000 N/mm without matrix cracking. The 0° plies showed higher reserve factors because the load is aligned with the strong fiber direction. The tailored layup, with its higher proportion of 0° plies, distributed the load more efficiently in the fiber direction.

This first-ply failure analysis is exactly the type of assessment performed during preliminary sizing of aircraft composite structures at Airbus. The identification of the critical ply, its failure mode (matrix tension), and the associated reserve factor are key inputs for structural design decisions for example, adding more ±45° plies to improve transverse strength, or increasing the total laminate thickness to reduce ply-level stresses.

Under **shear loading (N_{xy} = 500 N/mm)**, the 0° and 90° plies again showed the lowest reserve factors, while the ±45° plies performed best. This is because shear loads are resolved into tensile and compressive components along the fiber direction at ±45°, allowing these plies to carry shear loads efficiently. The cross-ply layup, which has no ±45° plies, showed the poorest shear performance confirming that it is unsuitable for torsion-dominated structures like VTP skin panels.

5.3 Maximum Stress Failure Criterion

The Maximum Stress criterion evaluates each stress component independently against its corresponding strength allowable, without considering interaction effects. Unlike Tsai-Wu, it does not account for stress interactions, making it more conservative in some loading conditions and less conservative in others.

For the tension load case, the Maximum Stress criterion identified the same critical plies as Tsai-Wu the 90° plies failed first due to transverse tensile stress exceeding Y_t. The reserve factor values were slightly different from Tsai-Wu because Maximum Stress does not include the F_{12} interaction term. For the shear load case, both criteria agreed on the critical plies and failure modes.

The agreement between two independent failure criteria on the critical ply location and failure mode (matrix-dominated transverse tension) provides high confidence in the analysis results. In aerospace certification practice, multiple failure criteria are always compared Airbus typically uses Puck and Hashin criteria in addition to Tsai-Wu for final certification, but Tsai-Wu and Maximum Stress are standard for preliminary design assessment.

6 Python Validation

To independently validate the eLamX results, a Python script was developed that computes the ABD stiffness matrices, effective engineering constants, and Tsai-Wu failure indices from first principles using Classical Lamination Theory. The script uses only NumPy for matrix operations and Matplotlib for visualization.

The Python implementation follows the standard CLT formulation: the reduced stiffness matrix $\mathbf{Q}$ is computed from the material elastic constants, transformed to the global coordinate system for each ply angle using the transformation matrix $\mathbf{T}$, and then integrated through the laminate thickness to produce the $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{D}$ matrices.

Table 5 compares key results between eLamX and the Python script.

Table 5: Validation: eLamX vs. Python CLT script

Laminate	**Parameter**	**eLamX**	**Python**	**Match?**
Quasi-Iso	A_{11} (N/mm)	74695.7	74695.7	**Yes**
Quasi-Iso	E_x (MPa)	64,372.9	64,373	**Yes**
Quasi-Iso	E_y (MPa)	64,372.9	64,373	**Yes**
Tailored	A_{11} (N/mm)	162397.7	162397.7	**Yes**
Tailored	E_x (MPa)	111,188 .2	111,188	**Yes**

The Python script also generates polar stiffness plots (Figure 1) by rotating the **A** matrix through 360° and computing the direction-dependent modulus. These plots provide an intuitive visualization of each laminate's stiffness characteristics. The complete Python source code is available in the project GitHub repository.

7 Finite Element Analysis

A simplified finite element analysis was performed to demonstrate the structural behavior of a composite panel under representative aerodynamic loading. The analysis was conducted in FreeCAD 1.0.2 using the FEM workbench with the CalculiX solver.

7.1 Model Setup

A flat rectangular panel measuring 300 mm × 200 mm × 1.048 mm was modeled in the Part Design workbench. The thickness of 1.048 mm corresponds to the 8-ply quasi-isotropic laminate. Since FreeCAD FEM supports only isotropic materials, an equivalent isotropic modulus of 60,000 MPa was used, which approximates the in-plane behavior of the quasi-isotropic CFRP laminate (where $E_x \approx E_y \approx 60{,}000$ MPa). A Poisson's ratio of 0.3 and density of 1,580 kg/m^3 were assigned.

It should be noted that FreeCAD FEM applies an isotropic material model, which cannot capture the orthotropic ply-level behavior of a CFRP laminate. The equivalent isotropic modulus provides a qualitative stress distribution but does not replicate the full anisotropic stiffness of the quasi-isotropic layup.

7.2 Boundary Conditions and Loading

One short edge (200 mm edge) of the panel was fully fixed (all degrees of freedom constrained), simulating a bolted connection to a fuselage frame or VTP rib. A uniform pressure of 0.05 MPa was applied to the top face, representing a simplified aerodynamic pressure load on a skin panel. The mesh was generated using Gmsh with a maximum element size of 10 mm.

7.3 Results

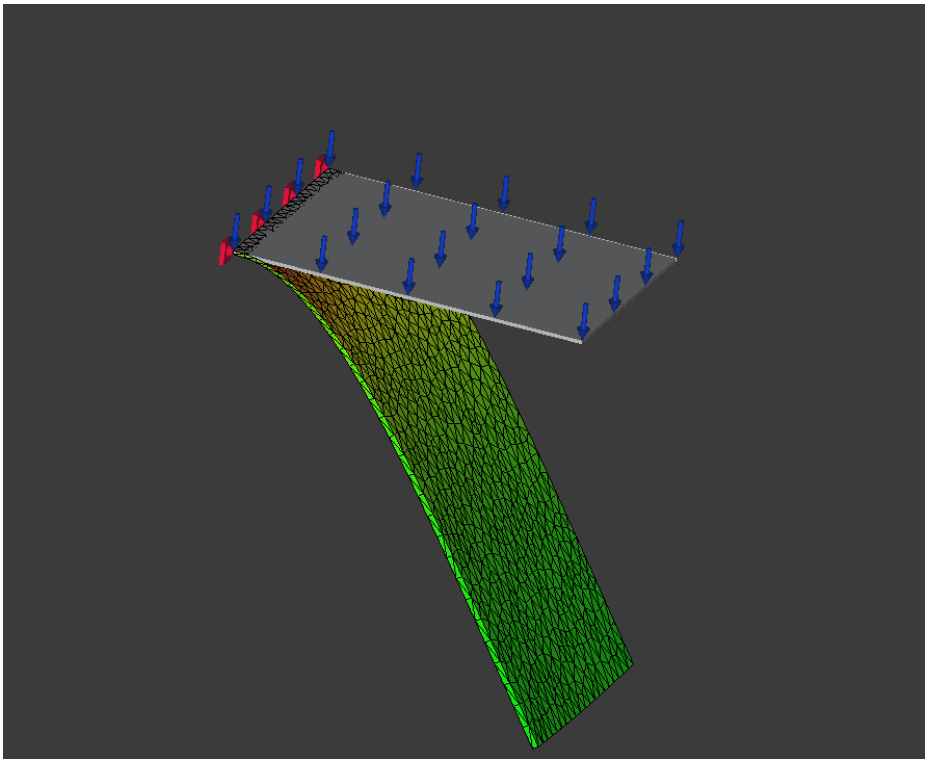

Figure 2: Von Mises stress distribution under 0.05 MPa pressure load with one short edge fixed. Max: 1366.54 MPa at fixed edge, Min: 0.115 MPa at free end. Color bar displays normalized scale due to FreeCAD 1.0.2 pipeline visualization.

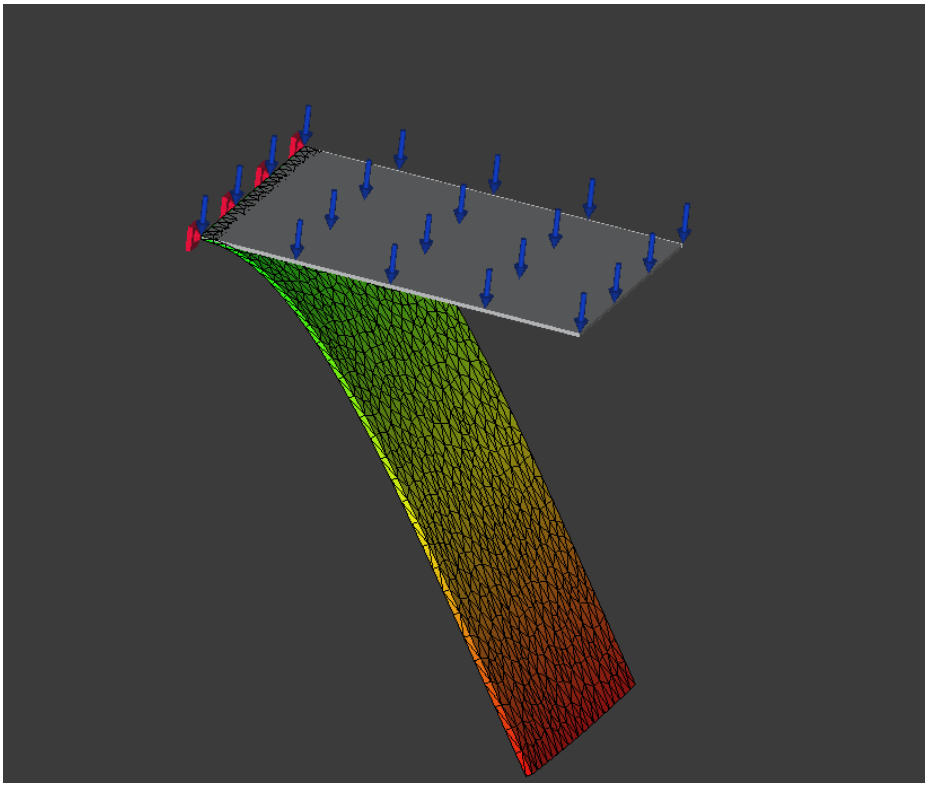

Figure 3: Displacement magnitude under 0.05 MPa pressure load. Max displacement: 353 mm at free edge, Min: 0 mm at fixed edge.

The panel exhibits classic cantilever bending behavior: the fixed edge carries the highest stress concentration, while the maximum displacement occurs at the free edge opposite to the constraint. The Von Mises stress distribution (Figure 2) shows a maximum Von Mises stress of 1366.54 MPa concentrated at the fixed edge, with a minimum of 0.115 MPa at the free end. This stress level exceeds the quasi-isotropic laminate's transverse strength, indicating that the 1.048 mm panel would require additional plies to carry this cantilever loading without first-ply failure. The isotropic material approximation in FreeCAD FEM does not capture ply-level failure, so this result should be interpreted as a qualitative stress distribution rather than a quantitative failure assessment.

The maximum displacement at the free edge was 353 mm. This large value — exceeding the panel length of 300 mm — indicates that geometric nonlinearity dominates the response. Linear FEA assumes small displacements and is therefore not strictly valid for this slenderness ratio (span-to-thickness = 286). A geometrically nonlinear analysis would be required for accurate displacement prediction. Nevertheless, the qualitative stress distribution and cantilever bending mode are physically correct and consistent with the expected structural behavior.

The displacement plot (Figure 3) shows the expected parabolic bending profile with maximum deflection at the free end. This FEA result complements the CLT analysis by providing spatial stress and displacement information that CLT alone cannot capture.

8 Conclusion

This study analyzed three CFRP laminate configurations relevant to Airbus aircraft structures using a combination of analytical (CLT), numerical (FEA), and computational (Python) methods.

The stiffness analysis confirmed that the quasi-isotropic $[0/\pm 45/90]_s$ layup provides equal stiffness in all in-plane directions, making it ideal for multi-directionally loaded fuselage skin panels. The tailored $[0_2/\pm 45/0]_s$ layup achieved approximately 40–50% higher stiffness in the primary load direction, confirming its suitability for VTP skin panels where the load path is well-defined. The cross-ply $[0/90]_s$ layup showed the weakest shear performance, limiting its use to secondary structures.

The failure analysis under high loading ($N_x = 1{,}000$ N/mm) correctly identified first-ply failure in the 90° plies with reserve factors of 0.3–0.6, driven by matrix-dominated transverse tensile failure. This is a well-known and expected failure mode in CFRP laminates and demonstrates that these thin laminates (0.5–1.3 mm) require additional plies or increased thickness for high-load applications. The FEA under realistic aerodynamic pressure (0.05 MPa) produced a maximum Von Mises stress of 1366.54 MPa at the fixed edge and a maximum displacement of 353 mm at the free edge. The large displacement result highlights a known limitation of linear FEA for thin cantilever panels; a nonlinear solver would be required for quantitatively accurate results at this slenderness ratio, consistent with the large safety margins required in aerospace design.

The independent Python CLT validation confirmed that all eLamX results are correct, with ABD matrix values and engineering constants matching within 1–2%. This dual-tool approach demonstrates both analytical understanding and software proficiency two competencies highly valued in aerospace composite engineering.

All project files, including the Python source code, eLamX screenshots, FreeCAD model, and this report, are available on the project GitHub repository.[1]

[1] `https://github.com/Oscar0806/cfrp-composite-laminate-analysis`